Algebra 2

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · FLORIDA COURSE 1200330 · FULL YEAR (TWO SEMESTERS)

~4¼ hrs per week
50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in the lessons, examples, and practice sets from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Algebra 2 extends the function families of Algebra 1 into a complete toolkit. The first semester opens the number system to complex numbers, masters quadratic equations and graphs, and moves into polynomial, rational, and radical equations and functions. The second semester develops rational exponents, logarithms, and exponential and logarithmic functions, then closes by analyzing functions as objects: properties, inverses, absolute value and piecewise functions, and systems that mix linear and nonlinear equations.

THE YEAR AT A GLANCE

- **M1 Square Roots & Complex Numbers** (~1½ wks) — imaginary numbers and operations
- **M2 Quadratic Equations in One Variable** (~2½ wks) — factoring, square roots, quadratic formula
- **M3 Quadratic Functions & Inequalities** (~2½ wks) — graphs, transformations, two-variable inequalities
- **M4 Polynomial Equations in One Variable** (~2½ wks) — operations, factoring, Remainder Theorem
- **M5 Polynomial Functions in Two Variables** (~1½ wks) — graphs and transformations
- **M6 Rational Expressions & Functions** (~2 wks) — equations, asymptotes, graphing
- **M7 Radical Equations & Functions** (~1½ wks) — solving, graphing, transformations
- **M8 Rational Exponents & Radical Expressions** (~1½ wks) — laws of exponents; the number e
- **M9 Logarithms & One-Variable Equations** (~1½ wks) — properties; exponential and log equations
- **M10 Exponential & Logarithmic Functions** (~2 wks) — graphs and transformations
- **M11 Properties of Functions** (~2 wks) — function types; even and odd; comparing
- **M12 Inverse Functions** (~1½ wks) — operations, composition, inverses
- **M13 Absolute Value & Piecewise Functions** (~2 wks) — equations, inequalities, graphs
- **M14 Systems of Equations & Inequalities** (~1½ wks) — linear and nonlinear systems

Honors. Honors scholars add a Probability module in Semester 1 and a Matrices module in Semester 2, plus extra lessons such as the Binomial Theorem, sequences, and piecewise functions.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- A review and module assessment as each module closes, roughly every one to two weeks.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific or graphing calculator for the exponential and logarithmic computations.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.